



Review Article

Geospatial Analysis of Air Pollution Sources and Their Impact on Public Health and the Environment in Western Uttar Pradesh, India

Bushra Praveen*

School of Environmental Studies, Maa Shakumbhari University, Punwarka, Saharanpur, Uttar Pradesh 247001, India

* Corresponding author. Email: bushraparveen8865@gmail.com

Abstract

Western Uttar Pradesh is among the most severely polluted regions of India, a condition driven by rapid urbanisation, dense vehicular traffic, an extensive small-scale manufacturing base and seasonal agricultural biomass burning. This review synthesises ambient air quality data for seven urban centres (Ghaziabad, Noida, Meerut, Agra, Firozabad, Saharanpur and Muzaffarnagar) from 2020 to 2026, covering the Air Quality Index (AQI) and six criteria pollutants. Particulate concentrations remain persistently above Central Pollution Control Board and World Health Organization reference values, with winter maxima and monsoon minima at every site. Two patterns run counter to the assumption of uniform deterioration: particulate loadings have fallen at most sites since the 2021–2022 peak, whereas ozone and nitrogen dioxide have risen sharply and now define the region's principal remaining risk. The particulate improvement is confined to the coarse fraction, so total mass has fallen while the respirable combustion fraction has not. The review also examines three processes that remain poorly characterised for the region: the possible role of airborne microplastic fibres in road and construction dust as carriers of adsorbed heavy metals; fugitive, non-stack emissions from informal micro, small and medium enterprise clusters, which chimney-based compliance monitoring cannot capture; and the mixing of agricultural residue smoke from the north-west with urban industrial plumes. Because no localised measurements of these processes exist for Western Uttar Pradesh, they are presented as evidence gaps and as a measurement agenda, not as quantified findings. Monitoring must be widened beyond the criteria-pollutant, single-city model towards regional airshed management.

ARTICLE HISTORY

Received: 23 June 2026
Revised: 09 August 2026
Accepted: 31 August 2026
Published: 15 September 2026

KEYWORDS

air pollution
air quality index
airborne microplastics
airshed management
fugitive emissions
particulate matter
western Uttar Pradesh

COPYRIGHT

© 2026 Author(s)
e-ISSN: 2583-942X

Exclusive Rights: Society for
AgroEnvironmental Sustainability,
Dehradun, India

LICENSE



This is an Open Access Article
published under a CC BY 4.0
International License

Citation: Praveen, B. (2026). Geospatial Analysis of Air Pollution Sources and Their Impact on Public Health and the Environment in Western Uttar Pradesh, India. *AgroEnvironmental Sustainability*, 4(3), 424–435. <https://doi.org/10.59983/s20260403019>

Statement of Sustainability: Improving air quality in Western Uttar Pradesh is essential for sustainable urban development, public health, ecosystem protection and climate resilience. Sustainable air-quality management should prioritise regional airshed-based planning, cleaner transport, energy-efficient industries, effective control of fugitive and non-stack emissions, and alternatives to agricultural residue burning. Strengthening continuous monitoring of particulate matter, ozone, nitrogen dioxide and emerging pollutants, including microplastic-associated contaminants, will support evidence-based interventions. Collaboration among government agencies, industries, agricultural communities, researchers and local stakeholders is necessary to reduce emissions equitably. Integrating pollution control with sustainable urban planning can deliver long-term environmental, social and economic benefits.

1. Introduction

Western Uttar Pradesh occupies the upper Ganga–Yamuna doab and functions as one of northern India's most important economic corridors. The region combines a dense industrial belt (the National Capital Region satellite cities of Ghaziabad and Noida, the engineering and sports-goods cluster of Meerut, the leather and footwear industry of Agra, the glass-bangle furnaces of Firozabad, the paper and pulp mills of Saharanpur and the sugar and jaggery units of Muzaffarnagar) with some of the most intensively cultivated agricultural land in the country. The combined population of these districts exceeds 75 million. This concentration of industry, agriculture and traffic within a single, topographically enclosed plain shapes air quality in three ways. Emission sources are unusually diverse, spanning combustion, process, mobile and area sources within short distances of one another. Much of the



industrial base consists of informal or semi-formal micro, small and medium enterprises whose emissions are neither metered nor reported. Over both, the Indo-Gangetic Plain experiences a persistent winter temperature inversion that suppresses the mixing height and traps pollutants close to the surface for weeks at a time (Dumka, 2025). These conditions make the region an airshed rather than a set of independent urban sources, yet regulatory monitoring and enforcement continue to operate on a city-by-city basis.

Ambient monitoring across the region between 2020 and 2026 records repeated and prolonged exceedance of national ambient air quality standards for particulate matter, a pattern documented across other Indian states over comparable periods (Ghosh et al., 2024). Annual mean PM_{10} concentrations at the more heavily loaded sites remain several times the CPCB annual limit of $60 \mu g/m^3$, and annual mean $PM_{2.5}$ concentrations remain roughly two to three times the CPCB annual limit of $40 \mu g/m^3$ throughout the record. Measured against the substantially stricter WHO 2021 global air quality guideline values of $15 \mu g/m^3$ for PM_{10} and $5 \mu g/m^3$ for $PM_{2.5}$ (World Health Organization, 2021), the exceedance factors are an order of magnitude larger. Because the health-relevant metric is cumulative exposure rather than peak concentration, sustained exceedance of this magnitude across a population of this size represents a chronic public health burden rather than an episodic pollution problem (Cohen et al., 2004; Künzli et al., 2000; Landrigan, 2017).

The principal contributing sources identified for the region are vehicular exhaust and road dust resuspension along the Delhi–Dehradun and Delhi–Agra corridors; combustion in brick kilns, glass furnaces and small foundries; construction and demolition activity in the rapidly expanding urban peripheries; open burning of municipal and agricultural waste; and the seasonal advection of crop residue smoke from Punjab and Haryana during the post-monsoon harvest window (Awasthi et al., 2024). The relative weight of these sources shifts through the year, which is why seasonal disaggregation is essential to any interpretation of the annual means.

Table 1. Mapping of prior research relevant to the Western Uttar Pradesh airshed.

Thematic area	Citation	Contribution to the regional evidence base
Regional air quality characterisation	Dumka (2025)	Spatio-temporal analysis of air pollution and meteorological influence in Western Uttar Pradesh using geospatial techniques; establishes the regional baseline for this study.
	Rajarithnam et al. (2014)	Measured air pollutant emissions from South Asian brick kilns, providing emission factors applicable to the region's kiln-dense districts.
Fugitive and MSME emissions	Weyant et al. (2014)	Measured emissions from thirteen South Asian brick kilns; emission factors differed between kiln types by up to an order of magnitude, with dominant traditional kilns the highest $PM_{2.5}$ emitters.
Airborne microplastics	Brahney et al. (2021)	Atmospheric transport modelling identifying microplastics as long-range vectors for co-adsorbed chemical hazards.
	Rao et al. (2025)	First characterisation of airborne microplastics in Delhi NCR; PET and polyethylene dominant, with zinc, silicon and aluminium adsorbed onto particle surfaces and weathering features observed.
	Pandey et al. (2022)	First evidence of microplastics in aerosols and street dust in Varanasi, Uttar Pradesh; FTIR polymer identification across sampling sites.
	Banerjee & Singh (2026)	FTIR-confirmed PE, LDPE, PP, PET and EVA in airborne dust at Agra; SEM revealed surface cracking and photo-oxidative features that can enhance sorption of co-pollutants.
	Singh et al. (2020)	Review of global atmospheric microplastic status with emphasis on Indian studies; finds research in India remains at an early stage with few reported studies.
Transboundary transport and source apportionment	Awasthi et al. (2024)	Source apportionment of 111 VOCs and PM in Delhi; differentiated fresh paddy-residue burning from other biomass sources, contributing 23–25% of PM_{10} and $PM_{2.5}$ in post-monsoon episodes.
	Nandi et al. (2026)	WRF-Chem modelling over Delhi–NCR: crop-residue-burning controls confined to Delhi–NCR reduce $PM_{2.5}$ by only 2–3%, while airshed-wide control yields a three- to four-fold larger benefit.
Health burden quantification	Pandey et al. (2021)	Global Burden of Disease based estimation of the health and economic impact of air pollution across Indian states.
	Kaur & Pandey (2021)	Review of the air pollution, climate change and human health nexus in Indian cities.

Beyond the criteria pollutants, three processes are repeatedly identified in the wider literature as material to airshed behaviour but remain unquantified for Western Uttar Pradesh specifically. Weathered synthetic polymer fibres are now recognised as a component of urban and peri-urban atmospheric dust. Oxidised functional groups on weathered fibre surfaces provide adsorption sites for metal ions, so the fibre may act as a physical carrier that delivers adsorbed lead, chromium or cadmium deeper into the respiratory tract than the free ionic species would reach, and retains it there longer. No published measurement of atmospheric microplastic abundance, polymer composition or associated metal loading exists for these districts.

Compliance monitoring in India is built around stack sampling at fixed chimney locations. Emissions that escape through low furnace openings, doorways, roof vents, material transfer points and open storage yards are structurally invisible to this design. Given that a large share of the region's glass, chemical, foundry and jaggery capacity sits in informal clusters with low-height or



absent stacks, the fraction of total industrial emission that is unaccounted for may be substantial, but it has not been estimated for the region. Post-monsoon residue burning smoke advected from the north-west arrives over the western districts already aged and chemically processed, and mixes with locally generated industrial and vehicular emissions. The resulting mixture is not simply additive: secondary organic and inorganic aerosol formation and ozone photochemistry depend on the ratio in which the two plumes combine. Source apportionment that separates the rural biomass and urban industrial contributions has been carried out for Delhi–NCR but not extended eastward into the Western Uttar Pradesh districts.

These three gaps share a common structural cause: the region's monitoring architecture measures the pollutants it was designed to regulate, at the locations where regulation is enforced. None of the three has been quantified for the region. Table 1 maps the principal strands of prior work, grouped by the aspect of the regional airshed each concerns.

2. Data and Methods

2.1. Study Area

Seven urban centres of Western Uttar Pradesh were selected for analysis: Ghaziabad, Noida, Meerut, Agra, Firozabad, Saharanpur and Muzaffarnagar. Selection was based on three criteria: the presence of a continuous ambient monitoring record for the full study period, representation of the region's distinct emission profiles, and administrative importance within the state's air quality management programme. The set spans the National Capital Region periphery (Ghaziabad, Noida), the central industrial belt (Meerut, Muzaffarnagar), the southern tourism and manufacturing corridor (Agra, Firozabad) and the northern agro-industrial zone (Saharanpur).

2.2. Data sources and Study Period

The analysis draws on secondary ambient air quality data for the period 2020 to 2026. Annual mean concentrations for PM₁₀, PM_{2.5}, SO₂, NO₂, O₃ and CO, together with annual mean AQI values, were compiled for each of the seven cities from publicly reported continuous ambient air quality monitoring records. Historical air-quality data for selected locations in Western Uttar Pradesh were obtained from AQI.in, an independent air-quality data platform. The platform provides air-quality observations and AQI information from monitoring networks and other sources. Where possible, monitoring-station information and pollutant observations were cross-referenced with data reported through the Central Pollution Control Board (CPCB) National Air Quality Index system. No primary sampling was undertaken. Two features of the record require explicit statement. The 2026 values represent a partial year and are therefore not strictly comparable with the six complete years that precede them; comparisons involving 2026 are treated as indicative throughout. Monitoring coverage is also uneven across the seven cities, since Ghaziabad and Noida are served by a denser network than Firozabad or Muzaffarnagar, so inter-city differences of small magnitude should not be over-interpreted.

2.3. Pollutants, Units and Reference Standards

Concentrations of PM₁₀, PM_{2.5}, SO₂, NO₂ and O₃ are reported in micrograms per cubic metre ($\mu\text{g}/\text{m}^3$). Carbon monoxide is conventionally reported in milligrams per cubic metre (mg/m^3) under the Indian National Ambient Air Quality Standards; the CO values compiled here are reported in the units of the source record and are used only for relative comparison across years and cities, not for direct comparison against the mg/m^3 standard. Observed concentrations are assessed against CPCB (2009) National Ambient Air Quality Standards and, where a stricter international benchmark is informative, against the WHO 2021 global air quality guidelines. The corresponding limit values are set out in Table 4, alongside the physiological mechanism associated with each pollutant. Air Quality Index values follow the CPCB National AQI scheme, in which the index is computed as the maximum of the sub-indices of the individual pollutants measured, and is classified as Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400) and Severe (401–500).

2.4. Analytical Approach

Three analyses were performed. Annual mean concentrations were tabulated by city and pollutant to establish the regional pollutant profile and to identify the direction of change over the study period (Table 2). Annual AQI series were then examined city by city to distinguish sites on an improving trajectory from those on a deteriorating one, and the highest and lowest monthly AQI values were extracted to characterise the seasonal amplitude at each site (Table 3). Finally, pollutant-specific behaviour was examined separately for the particulate and gaseous fractions, because the two groups display opposite trends over the study period and therefore imply different control priorities. Because the underlying data are annual and monthly aggregates rather than raw time series, the analysis is descriptive. No statistical testing, trend fitting or meteorological normalisation has been applied, and no causal attribution to individual emission sources is claimed. Mechanistic and epidemiological explanations of the observed patterns are drawn from the published literature rather than inferred from these aggregates.

3. Results



3.1. Regional Pollutant Profile, 2020–2026

Table 2 presents annual mean AQI and criteria pollutant concentrations for the seven cities across the full study period. Read across the region as a whole, the record does not describe uniform deterioration. It describes a divergence, in which the particulate fraction and the gaseous fraction have moved in opposite directions. Particulate loadings peaked early in the period and have since declined at every site. Regional PM₁₀ maxima occur in 2021 or 2022 and fall thereafter, in several cases substantially: Noida declines from 217 $\mu\text{g}/\text{m}^3$ in 2022 to 111 $\mu\text{g}/\text{m}^3$ in 2026, Ghaziabad from 233 $\mu\text{g}/\text{m}^3$ to 125 $\mu\text{g}/\text{m}^3$ over the same interval, and Firozabad from 225 $\mu\text{g}/\text{m}^3$ in 2020 to 85 $\mu\text{g}/\text{m}^3$ in 2026. PM_{2.5} shows the same early peak but a shallower decline, and at every site the 2026 value is higher than the period minimum, which falls in 2023 at five sites and in 2024 at Meerut. The implication is that the coarse fraction, dominated by road dust, construction activity and mechanically generated material, has responded to control measures, whereas the fine combustion-derived fraction has not. Gaseous pollutants show the reverse pattern. Ozone rises at all seven sites, in several cases steeply: Muzaffarnagar increases from 8 $\mu\text{g}/\text{m}^3$ in 2020 to 43 $\mu\text{g}/\text{m}^3$ in 2026, Meerut from 11 to 44 $\mu\text{g}/\text{m}^3$, and Noida from 9 to 30 $\mu\text{g}/\text{m}^3$. Nitrogen dioxide rises at five of the seven sites, with Ghaziabad increasing from 19 to 32 $\mu\text{g}/\text{m}^3$ and Muzaffarnagar from 7 to 24 $\mu\text{g}/\text{m}^3$; it is unchanged at Noida (13 $\mu\text{g}/\text{m}^3$ in both 2020 and 2026) and falls at Firozabad (23 to 12 $\mu\text{g}/\text{m}^3$). Because tropospheric ozone is a secondary pollutant formed photochemically from nitrogen oxides and volatile organic compounds, the parallel increase in both species is internally consistent and points to a growing precursor inventory, most plausibly the rising vehicle fleet, that is not being offset by the dust control measures responsible for the particulate improvement. Sulphur dioxide remains comparatively low in absolute terms across the region but rises sharply at two sites in 2026, reaching 21 $\mu\text{g}/\text{m}^3$ at Agra and 18 $\mu\text{g}/\text{m}^3$ at Firozabad against values of 4 to 9 $\mu\text{g}/\text{m}^3$ elsewhere. Both are locations with a concentrated furnace-based industrial base, and both warrant verification against the primary monitoring record before the increase is treated as established. The compiled carbon monoxide series shows a step change between 2021 and 2022 at every city simultaneously, which is characteristic of a change in instrumentation, reporting unit or averaging convention rather than of a regional emissions event; the CO column is therefore excluded from further interpretation.

Table 2. Annual mean AQI and criteria pollutant concentrations for seven cities of Western Uttar Pradesh, 2020–2026.

City	Pollutant	2020	2021	2022	2023	2024	2025	2026
Ghaziabad	AQI	192	229	238	187	204	193	173
	PM ₁₀	155	216	233	181	190	180	125
	PM _{2.5}	94	109	99	80	91	95	93
	NO ₂	19	19	21	20	20	25	32
	SO ₂	6	8	6	5	6	5	7
	O ₃	19	18	15	16	19	18	35
	CO	1	4	258	930	882	765	293
Noida	AQI	180	186	224	197	200	181	154
	PM ₁₀	126	160	217	195	186	165	111
	PM _{2.5}	97	94	92	84	87	87	90
	NO ₂	13	17	22	28	14	28	13
	SO ₂	4	4	6	5	6	4	4
	O ₃	9	13	14	15	19	22	30
	CO	0	0	233	1171	922	723	276
Meerut	AQI	175	196	192	177	153	158	167
	PM ₁₀	166	190	184	161	135	149	117
	PM _{2.5}	80	84	80	79	67	75	87
	NO ₂	16	39	17	16	14	18	25
	SO ₂	4	5	5	4	5	5	8
	O ₃	11	15	16	15	22	24	44
	CO	1	118	232	811	781	632	288
Muzaffarnagar	AQI	172	188	188	161	161	152	168
	PM ₁₀	160	178	173	141	139	137	112
	PM _{2.5}	79	81	84	72	74	72	85
	NO ₂	7	13	11	10	11	18	24
	SO ₂	6	7	7	5	6	6	9
	O ₃	8	12	19	17	20	22	43
	CO	1	218	169	622	646	673	301
Saharanpur	AQI	136	165	177	140	147	156	164
	PM ₁₀	103	148	163	125	131	138	110
	PM _{2.5}	68	73	79	64	67	76	86
	NO ₂	12	14	9	8	13	16	13
	SO ₂	3	5	6	5	6	5	6
	O ₃	19	15	17	12	19	20	27
	CO	1	230	174	530	687	680	278
Agra	AQI	141	151	121	82	93	110	131

Continued on next page...



Table 2. continued...

Firozabad	PM ₁₀	206	98	119	80	81	101	94
	PM _{2.5}	74	74	48	34	41	47	68
	NO ₂	15	12	12	9	13	10	16
	SO ₂	8	3	4	5	7	7	21
	O ₃	15	13	13	9	12	11	27
	CO	0	0	179	357	357	451	167
	AQI	275	135	120	87	84	89	121
	PM ₁₀	225	89	112	83	73	80	85
	PM _{2.5}	149	64	50	38	40	40	61
	NO ₂	23	9	13	9	7	8	12
	SO ₂	3	3	6	5	6	7	18
	O ₃	15	12	20	12	13	8	17
	CO	0	0	215	390	421	363	211

Note: PM₁₀, PM_{2.5}, NO₂, SO₂ and O₃ in $\mu\text{g}/\text{m}^3$; AQI dimensionless (CPCB National AQI). CO values are reproduced in the units of the source record and are not comparable with the CPCB mg/m^3 standard. 2026 values represent a partial year.

3.2. City-wise Annual AQI Trajectories

Annual mean AQI values conceal as much as they reveal, because the index is driven by whichever pollutant sub-index is highest in a given year and can therefore remain stable while the underlying pollutant mix changes. Read alongside Table 2, however, the AQI series usefully separates the seven cities into three trajectory groups. Table 3 summarises the trajectory and seasonal amplitude at each site.

Table 3. Annual AQI trajectory and seasonal amplitude by city, 2020–2026.

City	AQI 2020	AQI 2026	Period maximum (year)	Period minimum (year)	Highest monthly AQI	Lowest monthly AQI	Trajectory
Ghaziabad	192	173	238 (2022)	173 (2026)	464 (Nov 2021)	64 (Aug 2024)	Declining from 2022 peak
Noida	180	154	224 (2022)	154 (2026)	435 (Dec 2025)	55 (Jul 2024)	Declining from 2022 peak
Meerut	175	167	196 (2021)	153 (2024)	394 (Dec 2025)	41 (Jul 2023)	Rising since 2024
Muzaffarnagar	172	168	188 (2021, 2022)	152 (2025)	355 (Dec 2025)	Not reported	Rising since 2025
Saharanpur	136	164	177 (2022)	136 (2020)	379 (Dec 2025)	43 (Aug 2020)	Net deterioration over period
Agra	141	131	151 (2021)	82 (2023)	324 (Jan 2026)	Not reported	Rising since 2023
Firozabad	275	121	275 (2020)	84 (2024)	323 (Oct 2020)	36 (Aug 2024)	Large early gain, rising since 2024

Ghaziabad and Noida follow closely parallel paths. Both reach their highest annual AQI in 2022 and decline steadily thereafter, and in both cases the decline is driven predominantly by the coarse particulate fraction rather than by any broad improvement across pollutants. Both cities nonetheless retain the region's highest recorded monthly extremes, and in both the gaseous fraction is rising even as the index falls. The improvement in these two cities is therefore real but narrow, and the AQI understates the change in exposure character that lies beneath it.

Meerut, Agra, Firozabad and Muzaffarnagar share a common shape: a mid-period minimum followed by renewed increase in the final years of the record. Firozabad shows the most pronounced version of this pattern, falling from the highest annual AQI recorded anywhere in the dataset in 2020 to its minimum in 2024 before rising again; Agra reaches its minimum in 2023 and has risen in each subsequent year. In all four cities the recent increase coincides with rising nitrogen dioxide and ozone, and in Agra and Firozabad with a marked increase in sulphur dioxide. The mid-period minima at these sites should be interpreted cautiously, since the 2020 to 2021 interval includes the period of reduced industrial and transport activity associated with pandemic restrictions. Saharanpur is the only city in the set whose 2026 annual AQI exceeds its 2020 value, and its particulate and ozone concentrations both rise across the record. As the least industrialised of the seven cities at the start of the period, it is also the one where the trajectory is most plausibly attributable to growth in local emissions rather than to changes in regional background.

3.3. Seasonal Variation

The seasonal signal is the strongest single feature of the dataset and is consistent across all seven cities. At six of the seven sites the highest monthly AQI of the study period falls in November, December or January, and at every site for which a monthly minimum is reported the lowest value falls in July or August. The amplitude is large: at Ghaziabad the highest and lowest monthly values



differ by a factor of approximately seven, and comparable ratios apply at Noida, Meerut and Saharanpur. Two mechanisms account for this pattern. In winter, radiative cooling of the surface produces a shallow, stable boundary layer over the plain, reducing the volume into which emissions can disperse; wind speeds fall, and the post-monsoon agricultural residue burning season adds a large regional smoke burden on top of unchanged local emissions. In the monsoon, the reverse applies: convective mixing raises the boundary layer, and wet deposition removes both particles and soluble gases from the column. The seasonal cycle therefore reflects the interaction of meteorology with an emission inventory that is broadly constant through the year, rather than a seasonal change in emissions alone. It follows that annual mean values, while convenient for inter-year comparison, substantially understate the exposures that actually occur during the winter months. A same-window comparison of daily AQI values during July illustrates the point in the opposite direction. Within this monsoon window the highest daily values recorded across the study period range from 110 at Agra to 225 at Ghaziabad, and the lowest from 17 at Meerut to 30 at Ghaziabad. The entire monsoon range at every site sits below the winter monthly means at the same site. The July 2026 values fall in the Moderate band at most sites, which is the best air quality the region experiences at any point in the year.

3.4. Particulate Matter

The particulate record separates into two components that behave differently. PM_{10} declines at all seven sites from an early-period maximum, and the magnitude of the decline is largest where the initial loading was highest. This is the signature of source control acting on mechanically generated material: road dust suppression, construction site management, paving and mechanised sweeping all act preferentially on the coarse fraction, and their effect is proportionally greatest where uncontrolled dust dominated the baseline. $PM_{2.5}$ does not follow. Its regional decline is shallower, its minimum occurs later, and at all seven sites the 2026 value exceeds the mid-period minimum. The fine fraction derives principally from combustion: vehicle exhaust, brick kilns, glass and foundry furnaces, biomass and waste burning. None of these sources is affected by dust control. The consequence is a shift in the composition of the particulate burden: the total mass falls, but the fraction of that mass composed of respirable combustion particles, which carry the greater health risk per unit mass, rises. The $PM_{2.5}$ to PM_{10} ratio increases at every site over the study period. Assessments based on PM_{10} or on AQI alone will therefore record improvement in a region where the health-relevant exposure has changed character rather than diminished.

3.5. Gaseous Pollutants

Ozone increases at all seven sites and shows the largest proportional change of any pollutant in the dataset, with 2026 values at Meerut and Muzaffarnagar approximately four to five times their 2020 values. Nitrogen dioxide rises at five of the seven sites. Since ozone in the troposphere is not emitted directly but formed by the photochemical reaction of nitrogen oxides with volatile organic compounds under strong insolation, the concurrent increase in both species indicates a growing precursor inventory in a region that already receives high solar radiation for much of the year. This has a policy implication that runs against the intuition drawn from the particulate record. Reducing coarse dust lowers atmospheric turbidity and can therefore increase the actinic flux available to drive ozone photochemistry, so successful dust control may in part be accelerating ozone formation. Ozone also has a longer atmospheric lifetime than primary particulate matter and forms downwind of its precursor sources, which means it is inherently a regional rather than a municipal problem and cannot be managed within city boundaries. Sulphur dioxide remains low across most of the region, consistent with the substitution of coal by piped natural gas in parts of the industrial base and with the low sulphur content of current transport fuels. The exceptions are Agra and Firozabad, where the 2026 values are roughly two to five times those recorded elsewhere. Both cities host concentrated furnace-based industry, and the increase is large enough relative to the preceding years to warrant verification against the primary monitoring record before it is accepted as a real emissions change.

4. Health Implications

The health effects associated with each criteria pollutant are established in the toxicological and epidemiological literature (Anderson et al., 2012; Kelly & Fussell, 2015; 2012; Ghorani-Azam et al., 2016). Table 4 sets out, for each criteria pollutant, the applicable Indian standard, the physiological mechanism of injury and the principal clinical outcome associated with chronic exposure.

4.1. Respiratory Effects

The respiratory consequences of this exposure regime follow from particle size. Coarse particles are largely intercepted in the nose and upper throat, where they produce irritation, chronic cough and acute exacerbation in patients with pre-existing airway disease. Fine particles bypass this filtration entirely, reach the alveoli and provoke localised inflammation, with reduced lung function and chronic obstructive pulmonary disease as the long-term outcome (Kurt et al., 2016; Pope, 1989; 2000). Gaseous irritants act on the same tissue by a different route: sulphur dioxide dissolves in the airway lining fluid and provokes reflex bronchoconstriction within minutes, while nitrogen dioxide impairs the alveolar macrophage population responsible for clearing inhaled pathogens, raising susceptibility to bronchitis and pneumonia. Given that the gaseous fraction is the component rising across the region, the respiratory burden should be expected to shift over time from the dust-driven upper-airway presentation towards infection-related and ozone-related lower-airway disease.



Table 4. Criteria pollutants: Indian reference standards, mechanism of injury and principal health outcomes.

Pollutant	CPCB standard (residential/ industrial)	Mechanism of injury	Principal health outcomes of chronic exposure
PM ₁₀	100 µg/m ³ (24 h); 60 µg/m ³ (annual)	Deposition in the nasopharynx and upper airway; mucosal irritation	Throat irritation, chronic cough, acute exacerbation of asthma and bronchitis
PM _{2.5}	60 µg/m ³ (24 h); 40 µg/m ³ (annual)	Penetration to the alveoli, bypassing nasal filtration; translocation of ultrafine fraction across the alveolar–capillary barrier	Reduced lung function and COPD; systemic oxidative stress, arterial plaque progression, ischaemic heart disease and stroke
NO ₂	80 µg/m ³ (24 h); 40 µg/m ³ (annual)	Oxidative injury to airway epithelium; impairment of alveolar macrophage clearance	Increased susceptibility to lower respiratory infection, bronchitis and pneumonia; asthma exacerbation
SO ₂	80 µg/m ³ (24 h); 50 µg/m ³ (annual)	Rapid dissolution in airway lining fluid forming sulphurous acid; reflex bronchoconstriction	Bronchospasm and expiratory airflow limitation, most pronounced in asthmatic and hyper-responsive individuals
O ₃	100 µg/m ³ (8 h); 180 µg/m ³ (1 h)	Direct oxidative damage to the epithelial surface of the distal airway	Airway inflammation and pain on deep inspiration; progressive loss of lung capacity and interstitial scarring
CO	2 mg/m ³ (8 h); 4 mg/m ³ (1 h)	Binds haemoglobin with approximately 200 times the affinity of oxygen, forming carboxyhaemoglobin	Tissue hypoxia presenting as dizziness, confusion and headache; angina and cardiac decompensation in patients with existing coronary disease

Source: standards as specified in the CPCB National Ambient Air Quality Standards (Central Pollution Control Board [CPCB], 2009).

4.2. Cardiovascular and Systemic Effects

The ultrafine fraction of PM_{2.5} does not remain in the lung. Particles below approximately 100 nm cross the alveolar–capillary barrier and enter the systemic circulation, where they generate oxidative stress, promote endothelial dysfunction and accelerate the progression of arterial plaque (Nemmar et al., 2004; Geiser et al., 2005). The epidemiological consequence, established across a large international literature (Zmirou et al., 1998; Silva et al., 2016a; 2016b; Liu et al., 2018) and quantified for Indian states by Pandey et al. (2021), is elevated incidence of ischaemic heart disease, myocardial infarction and stroke in populations chronically exposed to elevated fine particulate concentrations. Because this pathway depends on the fine rather than the coarse fraction, the compositional shift already noted, falling total particulate mass with a rising fine fraction, implies that cardiovascular risk in the region may not be improving even where AQI values are falling.

Carbon monoxide operates by an unrelated mechanism, competing with oxygen for haemoglobin binding sites and reducing the oxygen-carrying capacity of the blood (Ghorani-Azam et al., 2016). The acute presentation is dizziness, confusion and headache; in patients with existing coronary disease, the reduction in oxygen delivery can precipitate angina.

4.3. Carcinogenic Risk

Particulate matter in the region carries adsorbed polycyclic aromatic hydrocarbons generated by incomplete combustion of fossil fuels, biomass and waste plastic (Gangwar et al., 2019; Cho et al., 2026). Several of these compounds are established human carcinogens, and long-term inhalation exposure is associated with increased rates of cellular mutation and elevated lung cancer risk (Turner et al., 2020). The relevant exposure metric is cumulative dose over years, which means that the sustained exceedance documented for the region, rather than any individual pollution episode, constitutes the carcinogenic hazard.

4.4. Vulnerable Population Groups

The regional burden is not distributed evenly. Children are disproportionately affected because alveolarisation and airway growth continue into adolescence: exposure during this window produces deficits in attained lung function that are not recovered later, and a growing literature associates early-life particulate exposure with adverse neurocognitive outcomes (Chu et al., 2026; Gordon et al., 2018). Children also breathe a greater volume of air per unit body mass than adults and spend more time outdoors. At the other end of the age distribution, older adults with established cardiorespiratory disease have reduced physiological reserve, so a given increment in exposure produces a larger clinical effect (Pope, 2000; Mannucci & Franchini, 2017). Outdoor workers, among them traffic police, street vendors, construction and brick kiln labourers, form a third group, exposed at concentrations well above the ambient monitored average for the greater part of the working day, and generally without occupational health surveillance.

4.5. Airborne Microplastics as a Co-exposure Pathway

This mechanism has a specific implication for the health assessment, and remains a hypothesis requiring testing rather than an established regional finding. If weathered synthetic polymer fibres are present in construction and road dust at the concentrations reported for other urban environments, they would constitute a distinct inhalation hazard for two reasons. Fibre geometry favours



deeper penetration and longer retention than equivalent-mass spherical particles, and the oxidised functional groups generated on weathered polymer surfaces provide binding sites for metal ions, so a fibre may deliver adsorbed lead, chromium or cadmium to the distal airway and retain it there (Rao et al., 2025; Brahney et al., 2021). The combined exposure would be neither the pure particulate exposure nor the pure metal exposure on which current risk assessments are based, and it would not be detected by monitoring that measures particulate mass alone. Whether this pathway is material in Western Uttar Pradesh cannot be determined from the available evidence, since no measurement of atmospheric microplastic abundance or associated metal loading has been reported for the region. Establishing it is the first research priority for the region.

5. Environmental and Agricultural Impacts

Western Uttar Pradesh is among the most productive agricultural regions in India, and the same atmospheric conditions that generate the regional health burden act directly on the crops grown beneath them. The agricultural cost of regional air pollution is therefore not a secondary consideration but a direct economic loss superimposed on the health loss (Sillmann et al., 2021).

5.1. Crop Yield and Photosynthesis

Ground-level ozone is the most phytotoxic of the regional pollutants, and the increase documented for the region therefore carries a direct agronomic cost. Ozone enters the leaf through the stomata during normal gas exchange and oxidises mesophyll cell membranes, producing visible foliar necrosis, reduced photosynthetic rate and lower assimilate partitioning to grain. Wheat is among the more ozone-sensitive of the major cereals, and the region's principal ozone season, when high insolation combines with elevated precursor concentrations, overlaps the wheat grain-filling period (Sillmann et al., 2021; Gupta et al., 2017). Sugarcane, the other dominant regional crop, is exposed for a longer fraction of its growth cycle. Because ozone forms downwind of its precursor sources, the agricultural areas between the urban centres may experience concentrations comparable to or higher than the monitored urban values (Cisneros et al., 2010).

Particulate deposition acts by a separate and more visible mechanism. Coarse dust and black carbon from brick kilns, unpaved roads and construction settle as a physical film on leaf surfaces, attenuating the photosynthetically active radiation reaching the chloroplast and, where deposition is heavy, occluding stomata (Van Zelm et al., 2016). The effect is greatest within a few hundred metres of the source, which in this region means the field margins along transport corridors and around kiln clusters. Unlike the ozone effect, this component of the burden should be expected to have eased over the study period, in line with the PM₁₀ decline recorded across the region.

5.2. Soil Acidification and Heavy-metal Loading

Sulphur and nitrogen oxides emitted from industrial corridors are oxidised in the atmosphere and returned to the surface as sulphuric and nitric acid, by wet deposition during the monsoon and by dry deposition through the remainder of the year. The cumulative effect is a downward drift in soil pH, mobilisation of aluminium into the soil solution, depletion of exchangeable base cations and suppression of the microbial communities responsible for nitrogen fixation and organic matter turnover (Cisneros et al., 2010). The consequence is a slow reduction in intrinsic soil fertility that is partially masked by increased fertiliser application, and which is therefore likely to be under-recognised. A second and less reversible process is the atmospheric deposition of heavy metals. Slag dust, foundry emissions and glass furnace particulate carry lead, chromium and cadmium. Sequential-extraction work in the Firozabad glass industrial area found cadmium, lead and copper to be strongly enriched by anthropogenic activity, with hazard quotients for nickel, manganese and chromium and excess lifetime cancer risk for Cr(VI) and nickel exceeding safe thresholds (Rajouriya et al., 2020). Once deposited these elements are not degraded; they partition between soil solution and exchange sites, are taken up by roots, and accumulate through the food chain. Given that the affected soils are under continuous intensive cultivation, the relevant time horizon for this contamination is decades, and the exposure route it creates, dietary rather than inhalational, falls entirely outside the scope of air quality monitoring.

6. Monitoring Blind Spots in the Regional Airshed

The monitored criteria pollutant record is the best available description of regional air quality, but its design determines what it can and cannot see, and two structural omissions are large enough to affect its interpretation.

6.1. Fugitive, Non-stack Emissions

Indian industrial air quality compliance is built around stack monitoring: emissions are sampled at a defined chimney outlet under defined conditions. This architecture measures the emissions that pass through the stack and, by construction, cannot measure those that do not. Emissions escaping from low furnace openings, doorways and roof vents, from material charging and transfer points, from open storage of raw material and slag, and from process leaks all fall outside it. The composition of the regional industrial base makes this omission consequential rather than marginal. The glass and bangle furnaces of Firozabad, the chemical and engineering units around Meerut and Ghaziabad, the jaggery and pulp units of Muzaffarnagar and Saharanpur, and the



foundry and leather clusters of Agra are dominated by micro, small and medium enterprises operating with low-height or absent stacks and limited process enclosure (Weyant et al., 2014; Rajouriya et al., 2020). Where such units cluster geographically, the aggregate fugitive release can plausibly exceed the aggregate stacked release, which would mean that the measured contribution of industry to regional loading is systematically understated. The magnitude of this unmeasured term remains a hypothesis; it has not been estimated for these districts, and quantifying it is a second research priority.

6.2. Transboundary Mixing of Rural and Urban Plumes

The second omission is spatial rather than technical. Monitoring in the region is organised by city, and each urban network is interpreted against its own local emission inventory. But the post-monsoon smoke plume advected from the agricultural north-west arrives over these cities already chemically aged, and mixes with locally generated industrial and vehicular emissions in proportions that vary with wind direction and boundary layer depth. The resulting mixture is not the arithmetic sum of its parts: secondary aerosol formation and ozone photochemistry both depend non-linearly on the ratio of the precursors supplied by each plume. The practical consequence is that a city-level monitoring station cannot distinguish the fraction of its measured loading that is locally generated and therefore locally controllable from the fraction that is imported. Real-time source apportionment capable of making this separation has been implemented for Delhi–NCR but not extended eastward across the Western Uttar Pradesh districts (Awasthi et al., 2024). Without it, municipal control measures are being designed against an emission inventory of unknown local share.

6.3. Implications for the Regulatory Model

Both omissions arise from the same underlying problem. The monitoring system measures the pollutants it was designed to regulate, at the locations where regulation is enforced, within the administrative boundaries in which enforcement is organised. Emissions that are diffuse in space, that occur outside a regulated outlet, or that cross a district boundary are not captured, not because they are unimportant, but because no part of the architecture is responsible for them. This is the structural argument for an airshed approach.

7. Limitations

Four limitations bound these conclusions. The analysis is entirely secondary: it compiles reported annual and monthly aggregates and undertakes no primary sampling, so it inherits any error present in the source record and cannot verify it. The aggregates are also descriptive rather than analytical, since no meteorological normalisation, trend fitting or statistical testing has been applied, which means that observed year-to-year differences cannot be separated into an emissions component and a weather component. Interannual variability in winter inversion strength alone is capable of producing changes of the magnitude reported here. A third constraint is the choice of start year. The study period opens in 2020, when industrial and transport activity were substantially reduced for part of the year. Any comparison that uses 2020 or 2021 as a baseline will therefore tend to overstate subsequent deterioration, and the mid-period minima observed at several sites should be read with this in mind. The 2026 values represent a partial year and are indicative only. Two features of the compiled record require qualification. The carbon monoxide series shows a simultaneous step change at all seven cities between 2021 and 2022 that is more consistent with a change in reporting unit or instrumentation than with a regional emissions event, and the reported magnitudes are not compatible with the mg/m^3 convention of the Indian standard; the CO column is accordingly excluded from interpretation. The 2026 sulphur dioxide values for Agra and Firozabad are two to five times those recorded at the other five cities and require confirmation. Monthly minima are not reported for Muzaffarnagar or Agra, so the seasonal amplitude at those two sites cannot be quantified. Finally, the microplastic, fugitive emission and transboundary mixing pathways are supported by mechanistic evidence from other regions but by no measurement from Western Uttar Pradesh; they remain hypotheses, and no quantitative claim is made about their contribution to regional loading.

8. Future Research Directions

These gaps translate into three measurable research programmes, each of which is feasible with existing instrumentation and each of which would close a specific hole in the regional evidence base. High-volume ambient samplers should be deployed at four contrasting site types: a high-traffic corridor, an active construction zone, an MSME cluster and a peri-urban agricultural control, with filter samples analysed by micro-FTIR spectroscopy to establish fibre abundance, polymer composition and size distribution. Paired elemental analysis of the same filters would establish whether adsorbed metal loading is detectable and how it varies with fibre weathering state. Sampling should span at least one full winter and one monsoon season, since the hypothesised increase in metal adsorption is expected to be greatest under the prolonged high-humidity, high-particulate conditions of winter smog episodes. Perimeter-based open-path FTIR along the boundaries of representative MSME clusters in Firozabad, Meerut, Saharanpur and Muzaffarnagar would allow the total flux leaving a cluster to be compared with the sum of its declared stack emissions, yielding a direct estimate of the fugitive fraction. Sensor-equipped unmanned aerial platforms would complement this by locating



individual release points within a cluster at a spatial resolution that fixed monitoring cannot achieve. The output required is a revised industrial emission inventory that carries an explicit fugitive term. Machine learning applied to the existing monitoring record, together with meteorological reanalysis and satellite fire-count data, could be used to forecast the conditions under which imported residue smoke and local urban emissions combine to produce the highest secondary pollutant concentrations (Zhu et al., 2026; He et al., 2026). The practical objective is a short-horizon operational forecast capable of supporting graded response actions, rather than a research model. Model development should be paired with real-time source apportionment so that forecast skill can be attributed to the correct source term.

9. Implementation and Policy Actions

The regional record supports a shift from municipal air quality management towards regional airshed management, a direction reinforced by the projected interaction of climate change with air pollution burdens (Kinney, 2018; Turnock et al., 2023). The case for that shift is not primarily administrative. It follows from three properties of the data: ozone, the fastest-rising pollutant in the region, forms downwind of its precursors and cannot be controlled within the boundary in which those precursors are emitted; a substantial share of winter particulate loading is imported across district boundaries; and the largest unmeasured emission term is diffuse and clustered rather than located at regulated outlets. None of these three can be tackled by a city acting alone; airshed-scale modelling for Delhi–NCR indicates that controls confined to a single administrative area deliver only a fraction of the benefit achievable across the full airshed. Table 5 shows staged programme, distinguishing measures that require no new institutional arrangement from those that do. The sequencing is deliberate: the immediate actions generate the measurements on which the medium-term programmes depend, and those programmes in turn generate the inventory on which regional enforcement would rest.

Table 5. Staged implementation plan for regional airshed management in Western Uttar Pradesh.

Horizon	Action	Responsible level	Verification indicator
Immediate (0–12 months)	Deploy low-cost sensor networks around high-density traffic junctions and MSME clusters in Ghaziabad, Noida, Meerut and Firozabad to identify unmonitored release points	District administration with UPPCB	Number of previously unmonitored release points identified and geolocated
Immediate (0–12 months)	Publish monthly and seasonal pollutant statistics alongside annual means, so that winter exposure is visible in the reported record	UPPCB / CPCB	Monthly AQI and pollutant series published for all seven cities
Immediate (0–12 months)	Issue graded winter advisories targeted at children, older adults and outdoor workers, timed to forecast inversion episodes	District health administration	Advisories issued in advance of episodes; occupational exposure guidance in force
Medium term (1–3 years)	Establish a high-volume ambient sampling and micro-FTIR programme to characterise atmospheric microplastic abundance, polymer composition and adsorbed metal loading	State universities and research institutions	Peer-reviewed regional baseline covering at least three contrasting site types
Medium term (1–3 years)	Conduct boundary-layer fugitive emission surveys of representative MSME clusters using open-path FTIR and sensor-equipped aerial platforms	UPPCB with technical partner	Revised industrial emission inventory incorporating an explicit fugitive term
Medium term (1–3 years)	Implement real-time source apportionment separating imported biomass smoke from local industrial and vehicular contributions	CPCB with research consortium	Apportioned source contributions reported for each of the seven cities
Long term (3–10 years)	Constitute Western Uttar Pradesh as a single statutory airshed management unit with a common emission inventory, monitoring standard and enforcement authority	State government / MoEFCC	Airshed authority established with binding regional reduction targets
Long term (3–10 years)	Direct clean air financing towards piped natural gas conversion and process enclosure in glass, foundry, pulp and jaggery clusters	State government with financing partner	Share of cluster capacity converted; measured reduction in fugitive release
Long term (3–10 years)	Build NO _x and VOC precursor control into regional planning, with protection of agricultural yield as an explicit stated objective	State government with department of agriculture	Reversal of the regional ozone trend; periodic crop loss assessment

10. Conclusion

The seven-city record does not describe a region in uniform decline, nor one in recovery. It describes a transition. Between 2020 and 2026 the coarse particulate burden fell at every site, and the annual AQI fell with it, an outcome attributable to dust control acting on mechanically generated material. Over the same period ozone rose at all seven sites and nitrogen dioxide at six, while the fine particulate fraction declined only shallowly and has risen again at several cities. Regional exposure has shifted from coarse and primary towards fine, secondary and gaseous, and the aggregate index used to report air quality is the metric least able to detect



that shift. The health burden should therefore not be assumed to be falling in proportion to the AQI, and the agricultural cost is rising, since ozone is both the fastest-increasing and the most phytotoxic pollutant in the dataset. Three further processes lie beyond what the present monitoring architecture can observe: weathered synthetic fibres as carriers of adsorbed metals, diffuse release from informal industrial clusters, and non-linear mixing of imported agricultural smoke with local plumes. None is quantified here. Closing these gaps requires ambient FTIR characterisation, perimeter fugitive flux surveys and real-time source apportionment.

References

- Anderson, J. O., Thundiyil, J. G., & Stolbach, A. (2012). Clearing the air: A review of the effects of particulate matter air pollution on human health. *Journal of Medical Toxicology*, 8(2), 166–175.
- Awasthi, A., Sinha, B., Hakkim, H., Mishra, S., Mummidivarapu, V., Singh, G., Ghude, S. D., Soni, V. K., Nigam, N., Sinha, V., & Rajeevan, M. N. (2024). Biomass-burning sources control ambient particulate matter, but traffic and industrial sources control volatile organic compound (VOC) emissions and secondary-pollutant formation during extreme pollution events in Delhi. *Atmospheric Chemistry and Physics*, 24, 10279–10304.
- Banerjee, D., & Singh, M. R. (2026). Airborne microplastic deposition at the Taj Mahal and Agra Fort, India. *Discover Environment*, 4(1), 105. <https://doi.org/10.1007/s44274-026-00611-9>
- Brahney, J., Mahowald, N., Prank, M., Cornwell, G., Klimont, Z., Matsui, H., & Prather, K. A. (2021). Constraining the atmospheric limb of the plastic cycle. *Proceedings of the National Academy of Sciences*, 118(16), e2020719118.
- Central Pollution Control Board. (2009). National Ambient Air Quality Standards. Notification in the Gazette of India, Ministry of Environment and Forests, New Delhi.
- Cho, A. K., Kumagai, Y., & Abiko, Y. (2026). The chemical toxicology of ambient air pollutants. *Annual Review of Pharmacology and Toxicology*, 66.
- Chu, W. K., Celis, T., Said, L., Hoet, P. H., & Vanoirbeek, J. A. (2026). Insights from human biomonitoring studies of children's exposure to indoor air pollutants in school and home environments: A systematic review. *Atmospheric Environment*, 121918.
- Cisneros, R., Bytnerowicz, A., Schweitzer, D., Zheng, S., Traina, S., & Bennett, D. H. (2010). Ozone, nitric acid, and ammonia air pollution is unhealthy for people and ecosystems in southern Sierra Nevada, California. *Environmental Pollution*, 158(10), 3261–3271.
- Cohen, A. J., Anderson, H. R., Ostro, B., Pandey, K. D., Krzyzanowski, M., Künzli, N., ... Smith, K. R. (2004). Urban air pollution. In *Comparative quantification of health risks: Global and regional burden of disease attributable to selected major risk factors* (Vol. 2, pp. 1353–1433).
- Dumka, U. C. (2025). Spatio-temporal analysis of air pollution and meteorological influences in western Uttar Pradesh using geospatial techniques: Insights for policy and management. *International Journal of Remote Sensing*, 1–28.
- Gangwar, C., Chaudhary, R., Chauhan, A., Kumar, A., Singh, A., & Tripathi, A. (2019). Assessment of air pollution caused by illegal e-waste burning to evaluate the human health risk. *Environment International*, 125, 191–199.
- Geiser, M., Rothen-Rutishauser, B., Kapp, N., Schürch, S., Kreyling, W., Schulz, H., Semmler, M., Im Hof, V., Heyder, J., & Gehr, P. (2005). Ultrafine particles cross cellular membranes by nonphagocytic mechanisms in lungs and in cultured cells. *Environmental Health Perspectives*, 113(11), 1555–1560.
- Ghorani-Azam, A., Riahi-Zanjani, B., & Balali-Mood, M. (2016). Effects of air pollution on human health and practical measures for prevention in Iran. *Journal of Research in Medical Sciences*, 21, 65.
- Ghosh, B., Barman, H. C., Ghosh, S., Habib, M. M., Mahato, J., Dayal, L., ... Padhy, P. K. (2024). Air pollution status and attributable health effects across the state of West Bengal, India, during 2016–2021. *Environmental Monitoring and Assessment*, 196(2), 165.
- Gordon, T., Ramakrishna, K., Dey, S., Rajagopalan, S., Thornburg, J., Thurston, G., ... Nadadur, S. (2018). Air pollution health research priorities for India: Perspectives of the Indo-US Communities of Researchers. *Environment International*, 119, 100.
- Gupta, R., Somanathan, E., & Dey, S. (2017). Global warming and local air pollution have reduced wheat yields in India. *Climatic Change*, 140(3), 593–604.
- He, Y., Jiang, Y., Cao, Q., Liu, Z., Han, J., Li, W., & Chen, Z. (2026). A study on the prediction and driving mechanisms of typical air pollutant concentrations in the Yunnan Plateau region using machine learning. *Atmospheric Environment*, 121880.
- Kaur, R., & Pandey, P. (2021). Air pollution, climate change, and human health in Indian cities: A brief review. *Frontiers in Sustainable Cities*, 3, 705131.
- Kelly, F. J., & Fussell, J. C. (2015). Air pollution and public health: Emerging hazards and improved understanding of risk. *Environmental Geochemistry and Health*, 37(4), 631–649.
- Kelly, F. J., Fuller, G. W., Walton, H. A., & Fussell, J. C. (2012). Monitoring air pollution: Use of early warning systems for public health. *Respirology*, 17(1), 7–19.
- Kinney, P. L. (2018). Interactions of climate change, air pollution, and human health. *Current Environmental Health Reports*, 5(1), 179–186.
- Künzli, N., Kaiser, R., Medina, S., Studnicka, M., Chanel, O., Filliger, P., ... Sommer, H. (2000). Public-health impact of outdoor and traffic-related air pollution: A European assessment. *The Lancet*, 356(9232), 795–801.
- Kurt, O. K., Zhang, J., & Pinkerton, K. E. (2016). Pulmonary health effects of air pollution. *Current Opinion in Pulmonary Medicine*, 22(2), 138.
- Landrigan, P. J. (2017). Air pollution and health. *The Lancet Public Health*, 2(1), e4–e5.
- Liu, W., Xu, Z., & Yang, T. (2018). Health effects of air pollution in China. *International Journal of Environmental Research and Public Health*, 15(7), 1471.
- Mannucci, P. M., & Franchini, M. (2017). Health effects of ambient air pollution in developing countries. *International Journal of Environmental Research and Public Health*, 14(9), 1048.
- Nandi, I., Ganguly, D., Habib, G., & Dey, S. (2026). Delhi cannot clean its air alone: Airshed-scale mitigation outperforms local controls even under unfavourable winter meteorology. *npj Clean Air*, 2(1), 27. <https://doi.org/10.1038/s44407-026-00065-6>
- Nemmar, A., Hoylaerts, M. F., Hoet, P. H. M., & Nemery, B. (2004). Possible mechanisms of the cardiovascular effects of inhaled particles: Systemic translocation and prothrombotic effects. *Toxicology Letters*, 149(1–3), 243–253.



- Pandey, A., Brauer, M., Cropper, M. L., Balakrishnan, K., Mathur, P., Dey, S., ... Dandona, L. (2021). Health and economic impact of air pollution in the states of India: The Global Burden of Disease Study 2019. *The Lancet Planetary Health*, 5(1), e25–e38.
- Pandey, D., Banerjee, T., Badola, N., & Chauhan, J. S. (2022). Evidences of microplastics in aerosols and street dust: A case study of Varanasi City, India. *Environmental Science and Pollution Research*, 29(54), 82006–82013.
- Pope, C. A., III. (1989). Respiratory disease associated with community air pollution and a steel mill, Utah Valley. *American Journal of Public Health*, 79(5), 623–628.
- Pope, C. A., III. (2000). Epidemiology of fine particulate air pollution and human health: Biologic mechanisms and who's at risk? *Environmental Health Perspectives*, 108(Suppl. 4), 713.
- Rajarithnam, U., Athalye, V., Ragavan, S., Maithel, S., Lalchandani, D., Kumar, S., Baum, E., Weyant, C., & Bond, T. (2014). Assessment of air pollutant emissions from brick kilns. *Atmospheric Environment*, 98, 549–553.
- Rajouriya, K., Rohra, H., & Taneja, A. (2020). Levels of fine particulate matter bound trace metals in air of glass industrial area, Firozabad. *Pollution*, 6(3), 621–638.
- Rao, M. N., Ghude, S. D., Nivdange, S. D., Panchang, R., Pipal, A. S., Mukherjee, A., Sharma, H., & Kumar, V. (2025). Characterization and health risk assessment of airborne microplastics in Delhi NCR. *Scientific Reports*, 15, 25662.
- Sillmann, J., Aunan, K., Emberson, L., Büker, P., Van Oort, B., O'Neill, C., ... Brisebois, A. (2021). Combined impacts of climate and air pollution on human health and agricultural productivity. *Environmental Research Letters*, 16(9), 093004.
- Silva, R. A., Adelman, Z., Fry, M. M., & West, J. J. (2016a). The impact of individual anthropogenic emissions sectors on the global burden of human mortality due to ambient air pollution. *Environmental Health Perspectives*, 124(11), 1776.
- Silva, R. A., West, J. J., Lamarque, J. F., Shindell, D. T., Collins, W. J., Dalsoren, S., ... Zeng, G. (2016b). The effect of future ambient air pollution on human premature mortality to 2100 using output from the ACCMIP model ensemble. *Atmospheric Chemistry and Physics*, 16(15), 9847–9862.
- Singh, N., Singh, S., & Mall, R. K. (2020). Urban ecology and human health: Implications of urban heat island, air pollution and climate change nexus. In *Urban ecology* (pp. 317–334). Elsevier.
- Turner, M. C., Andersen, Z. J., Baccarelli, A., Diver, W. R., Gapstur, S. M., Pope, C. A., III, ... Cohen, A. (2020). Outdoor air pollution and cancer: An overview of the current evidence and public health recommendations. *CA: A Cancer Journal for Clinicians*, 70(6), 460–479.
- Turnock, S. T., Reddington, C. L., West, J. J., & O'Connor, F. M. (2023). The air pollution human health burden in different future scenarios that involve the mitigation of near-term climate forcers, climate and land use. *GeoHealth*, 7(8), e2023GH000812.
- Van Zelm, R., Preiss, P., van Goethem, T., Van Dingenen, R., & Huijbregts, M. (2016). Regionalized life cycle impact assessment of air pollution on the global scale: Damage to human health and vegetation. *Atmospheric Environment*, 134, 129–137.
- Weyant, C., Athalye, V., Ragavan, S., Rajarithnam, U., Lalchandani, D., Maithel, S., Baum, E., & Bond, T. C. (2014). Emissions from South Asian brick production. *Environmental Science & Technology*, 48(11), 6477–6483.
- World Health Organization. (2021). WHO global air quality guidelines: Particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Geneva: World Health Organization.
- Zhu, X., Li, B., Cao, Y., & Zhang, Q. (2026). Applicability analysis of tree-based ensemble learning for air pollutant prediction models. *Scientific Reports*, 16(1), 9602.
- Zmirou, D., Schwartz, J., Saez, M., Zanobetti, A., Wojtyniak, B., Touloumi, G., ... Anderson, H. R. (1998). Time-series analysis of air pollution and cause-specific mortality. *Epidemiology*, 495–503.

Publisher's note/Disclaimer: Regarding jurisdictional assertions in published maps and institutional affiliations, We maintains its neutral position. All publications' statements, opinions, and information are the sole responsibility of their respective author(s) and contributor(s), not We or the editor(s). We and/or the editor(s) expressly disclaim liability for any harm to persons or property caused by the use of any ideas, methodologies, or products described in the content.